HEALTHCARE PRESENTATION · MACHINE LEARNING PROJECT

# Medical Insurance Presentation

Predicting annual medical insurance charges from age, BMI, smoking habits and other health-related details.

Group -32

**GitHub** github.com/Harsh-IIT-Ropar/insurance-cost-prediction

## Team & roles


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Documentation Lead

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Video Lead

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Execution Lead

# 2 Problem and data

A realistic synthetic healthcare insurance dataset for regression

## What and why

A realistic synthetic healthcare insurance dataset created for machine learning, data science, healthcare analytics and AI-based prediction projects. It contains demographic, lifestyle, medical history, chronic disease, hospitalization and insurance-related information of individuals along with their annual medical costs.

## Designed for

- **Medical Insurance Cost Prediction**
- **Healthcare Analytics**
- **Risk Assessment Systems**
- **Regression ML Projects**
- **Predictive Modeling**
- **AI Healthcare Dashboards**
- **Data Visualization & EDA**
- **Feature Engineering Practice**

**7,500**
rows

**22**
columns

TARGET VARIABLE

**annual_medical_cost_usd**

Dataset type **Structured Tabular Data**

ML problem type **Regression**

# 3 Approach

Project pipeline: dataset → model → prediction

1. **Dataset Collection**
   Medical insurance dataset
2. **Data Preprocessing**
   Cleaning and encoding data
3. **Model Training**
   Linear Regression
4. **Model Testing**
   Compare predicted and actual costs
5. **Final Prediction**
   Estimated annual insurance cost

## Approach Used – Linear Regression

Linear Regression is a supervised Machine Learning algorithm used to predict continuous numerical values.

In our project, it predicts annual medical insurance charges based on different input factors such as age, BMI, smoking habits and other health-related details.

## Libraries used

1. **numpy**
2. **pandas**
3. **matplotlib**
4. **scikit-learn**
5. **seaborn**

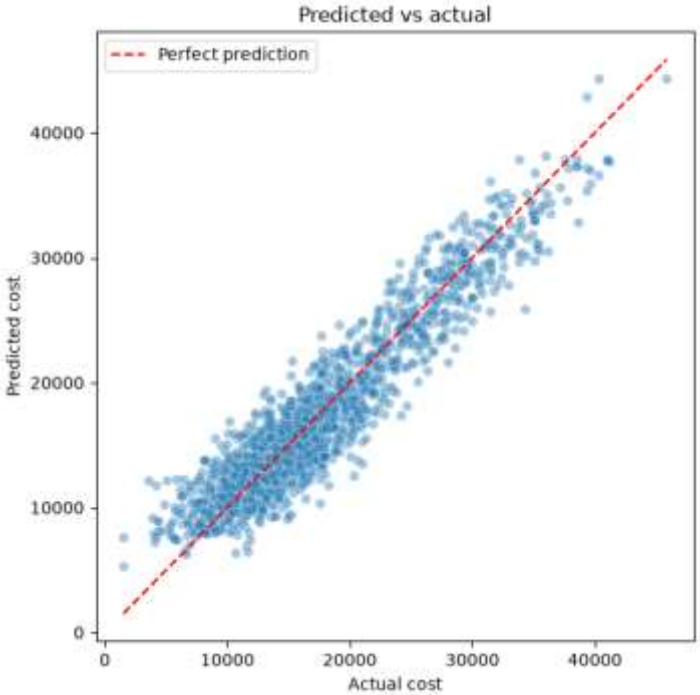
# 4 Results

After training and testing, the model predicts medical insurance costs from the input information

| $R^2$ score | RMSE | MAE |
|---|---|---|
| **0.8852** | **2516.85** | **2019.69** |
| test set | USD | USD |

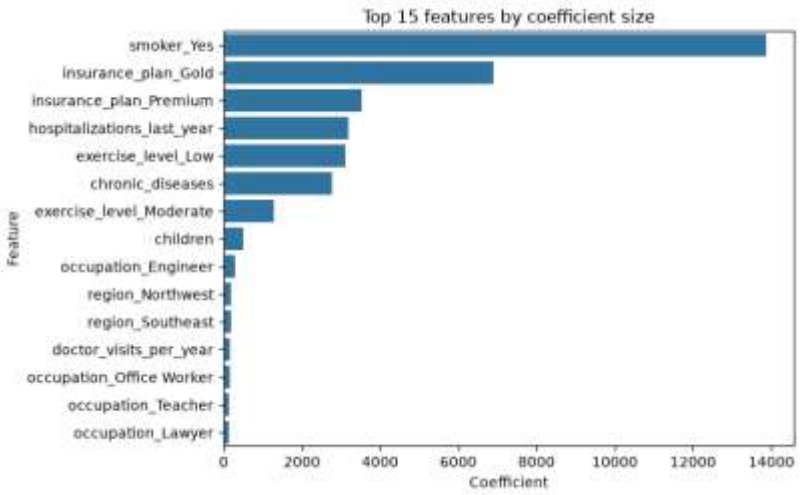
## Important performance metrics

- The model explains about 88.5% of the variation in annual medical cost on the test set, with a typical error of about $2,000 to $2,500.
- Smoking status had the largest effect: holding other factors equal, smokers had a predicted cost about $13,900 higher than non-smokers.
- Chronic diseases, hospitalizations, the Gold insurance plan and low exercise level were the next strongest factors, while region and occupation had little effect.



# 5 Learnings

What we took away and how AI was used




Smoking status dominates; region and occupation barely matter.

## What worked

- Linear Regression successfully predicts numerical insurance costs.
- Personal and lifestyle information helps estimate insurance expenses.
- Pandas made data processing easier.
- Matplotlib helped visualize the results.

## What didn't work

- Prediction errors may occur because of differences between actual and predicted values.
- Linear Regression may not capture all complex relationships in medical insurance costs.
- Missing or incorrect data can affect model performance.

**AI-use disclosure** AI tools were used to support project understanding, code assistance and presentation preparation. The dataset processing, model implementation and result analysis were performed as part of our project work.